

APSMO 2012 Set 2

1 Problems

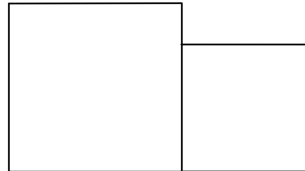
Problem 1.1 The sum of the digits of the number 2010 is 3.
What is the next larger number whose digit-sum is also 3?

Problem 1.2 In the pattern below, what will the 78th letter?

ABBCCDABBCCDA...

Problem 1.3 If you triple Jen's age and subtract 16, the result will be the same as when you double her age and add 8.
How old is Jen?

Problem 1.4 This figure consists of two squares. Each side length is a whole number of centimetres. The combined areas of the squares is 100cm^2 .
What is the perimeter of the entire figure?



Problem 1.5 A pile of coins sits on a table. Sara takes half of the coins plus 4 more. Then Nick takes 2. Then Joe takes 2 more than half of what is left. Finally Selena takes 5. Four coins remain on the table.
How many coins were on the table at the start?